

Mapping-Space Shear Initiality

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Summary

Pass 87 upgrades the finite-adele shear-pushout universal property to a derived mapping-space statement. For a shear-marked target

$$M = (0 \rightarrow D \rightarrow E \rightarrow \epsilon \rightarrow 0),$$

the restriction map

$$\mathrm{Map}(C_{\mathbb{Q}}, M) \rightarrow \mathrm{Map}(C_{\mathbb{Z}}, M)$$

has homotopy fiber

$$\mathrm{RMap}(\mathbb{Q}/\mathbb{Z}, D),$$

because the cofiber of $\mathbb{Z} \rightarrow \mathbb{Q}$ is $\mathbb{Q}/\mathbb{Z}$.

Main Claims

- **Fiber formula.** The obstruction to extending a shear-marked map from $C_{\mathbb{Z}}$ to $C_{\mathbb{Q}}$ is exactly $\mathrm{RMap}(\mathbb{Q}/\mathbb{Z}, D)$.
- **Contractible uniquely divisible case.** If D is uniquely divisible, then $\mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, D) = 0$ and higher Ext obstructions vanish, so the fiber is contractible.
- **Torsion decoration.** If D has torsion-divisible summand T , then $\mathrm{RMap}(\mathbb{Q}/\mathbb{Z}, T)$ contributes extra components. Strict initiality either excludes T or decorates the model by choosing this boundary component.

Machine Verification

`code/scripts/check-pass87.py` produced `artifacts/reports/pass87-mapping-space-shear-initiality-check.json` with overall PASS. It records the cofiber/fiber sequence, verifies contractible finite torsion tests for $\mathbb{Q}$ -vector kernels, and shows that a rank- r torsion-divisible finite shadow has N^r boundary components at modulus N .

Next Use

Pass 88 should compute the derived automorphism/stabilizer of the final finite-adele shear extension and compare it with the solid Borel $\mathbb{Q}^{\times} \ltimes \epsilon$.